

Impact of Stock Trading Apps on Indian Millennial Consumer Behavior in the Stock Market

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The present study is conducted on the ‘stock trading’ business and how consumers decide to buy and sell in the digital era. Currently, many apps are available on smartphones, such as ‘kite’ by Zerodha, Angel Broking Stock Trading App, ‘MO investor’ by Motilal Oswal, ‘IIFL Markets’ by IIFL securities, and many others which have more than ten lakhs of downloads on the Google Playstore. Lakhs of Indians, mostly millennials between the ages of 20 and 35, trade daily through these. Before the rise of these apps in the mid-2010s, people used to rely on their stockbrokers to place an order to sell or buy shares of a company. Information sources were just television and newspaper – and most people used to act on the advice of stockbrokers or their trusted acquaintances. Now, the scenario is completely different – stock trading individuals are continuously updated through their ‘Trading apps’ (such as Kite and others mentioned above) or are advised by gurus through social media apps like Instagram or LinkedIn or YouTube, etc. There are even Stocking Trading Advisory apps such as ‘Upstox’, ‘Smallcase’ etc., which advise consumers on purchase, sell, and hold decisions. This paper will focus on the impact of stock trading apps on the Indian Millennial (20 to 35 years) consumer behavior in the stock market.

This study will also focus on identifying the ‘main parameters of value’ the customers consider when deciding to engage in online trading through stock trading apps. This study will further undertake a competitive analysis of the discovered ‘main parameters of value’ in the most used apps, limiting to Zerodha, Angel Broking, Motilal Oswal, and IIFL securities, etc., which have higher than ten lakhs downloads in google playstore. This study will conclude by identifying which App amongst these is most ahead in its journey to becoming an ideal product at this time.

Keywords— Stock Market, Stock trading Apps, Millennial Consumer Behavior, Online Stock Trading

I. INTRODUCTION

Amid the covid-19 pandemic, ‘work’ has shifted from in-person meetings and offices to online meetings and everyone’s home. This digital shift has been occurring in all walks of life, not just work. That is why, when the pandemic hit, it was relatively easy to shift most of the jobs to online made. Mostly India’s Millennial population has been exposed to digital information for many years, but it was increased a lot due to lockdown.

The present study is conducted on the ‘stock trading’ business and how consumers buy and sell in the digital era. Currently, there are many apps available on a smartphone such as ‘kite’ by Zerodha, Angel Broking Stock Trading App, ‘MO

investor’ by Motilal Oswal, ‘IIFL Markets’ by IIFL securities, and many others which have more than ten lakhs of downloads on the Google Playstore. Lakhs of Indians, mostly millennials between the ages of 20 and 35, trade daily through these. Before the rise of these apps in the mid-2010s, people used to rely on their stockbrokers to put ‘an’ order to sell or buy any share of a company. Information sources were just television and newspaper – and most people used to act on the advice of stockbrokers or their trusted acquaintances. Now, the scenario is completely different – stock trading individuals are continuously updated through their ‘Trading apps’ (such as Kite and others mentioned above) or are advised by gurus through social media apps like Instagram or LinkedIn or YouTube, etc. There are even Stocking Trading Advisory apps such as ‘Upstox’, ‘Smallcase’ etc., which advise consumers on purchase, sell, and hold decisions. This paper will focus on the impact of stock trading apps on the Indian Millennial (20 to 35 years) consumer behavior in the stock market.

II. PROBLEM STATEMENT

It has been observed that in the past, consumers had to mainly depend on the stockbroker – an individual present at a physical location for trading. Most of the time, there used to be a lack of updated knowledge and information about companies, along with the chaos caused due to the gathering of a considerable number of consumers – consumers had very little control. Now, using stock trading apps, the consumer has complete control over the trading actions and complete access to all the relevant information and advice. This study will answer the question – ‘has this control and unlimited access resulted in an increased number of consumers in the millennial age bracket, also more trading action per individual, more investment per individual?’ when compared with millennials before the advent of smartphone stock trading, i.e., in the mid-2000s. This study will be done through ‘time series’ by comparing individuals in the same age and income groups across the periods (Money will be adjusted for inflation).

This study will also focus on identifying the ‘main parameters of value’ the customers consider when deciding to engage in online trading through stock trading apps. This study will further undertake a competitive analysis of the discovered ‘main parameters of value’ in the most used apps, limiting to Zerodha, Angel Broking, Motilal Oswal, IIFL securities, etc., which have higher than ten lakhs downloads in google playstore. This study will draw suggestions by identifying the App that is on its journey to becoming an ideal product at this time.

III. RESEARCH OBJECTIVES

- To compare the consumer behavior of Indian millennials using stock trading apps and offline stock trading
- To study the main parameters of value considered by consumers when choosing to engage in stock trading apps (exploratory research)
- To study and analyze the progress of stock trading apps – namely Zerodha, Angel Broking, Motilal Oswal, IIFL securities, GROWW, etc. on their continuous journey to becoming ‘ideal products’ for their consumers (competitive analysis)

IV. RESEARCH HYPOTHESIS

1. **H0:** There is no significant difference in the number of Indian Millennials consumers of the stock market using online trading apps, websites, and offline
H1: There is a significant difference in the number of Indian Millennials consumers of the stock market using online trading apps, websites, and offline
2. **H0:** There is no significant difference in the number of trading actions performed per consumer between trading app consumers, website consumers, and offline consumers
H1: There is a significant difference in the number of trading actions performed per consumer between trading app consumers, website consumers, and offline consumers
3. **H0:** There is no significant difference in the amount of money invested per consumer between trading app consumers, website consumers, and offline consumers
H1: There is a significant difference in the amount of money invested per consumer between trading app consumers, website consumers, and offline consumers

V. LITERATURE REVIEW

There have been a few research projects in the area of Indian consumer behavior in the Online stock market [1], which examined the consumer's preference for online and offline stock trading and compared their usages in 2007. They found that, in 2007, only 28% of the traders traded online. Online Stock trading did not account for more than half a percent of the total traded volume 14 years back. These results concluded that most Indian Investors are conservative, do not change quickly, and prefer offline stockbrokers.

In 2014, it was studied that the result of the growth of Internet connectivity in India on online stock trading [1]. She found that removing the major barrier of low internet speeds would result in a major shift in the number of people preferring offline trading to online trading.

[2] in 2017, conducted a study to identify factors that influence people to engage in online stock trading. Various factors such as internet skills, social and cultural perceived advantages, attitudes towards adopting and intending to use the internet were analyzed. This research was conducted online via mailed questionnaires [3]. The responses analyzed in SPSS found that age, gender, monthly income, and educational qualification have no significance in the intention to trade in the online stock platform.

Further, another empirical study was undertaken by [4] on the online trading system in the Indian stock market to study the perception of people [5] in Faridabad city. Using the ANOVA technique, they studied the growth of Online trading across demographics – including age, gender, occupation, and annual income. It was found that there is a significant difference across gender and annual income, where male and higher-income people prefer online stock trade more compared to female and lower-income people. It was also found that there is no significant difference across age and occupation.

[6], [7] had in 2015, conducted research on social media websites related to stock markets and the online stock market. They concluded that online stock trading has increased and expanded since its inception. There are many joint ventures from foreign investors and companies planning to enter this market in India. The costs and expenditures have been lowered significantly, and at the same time, the value of the Indian Stock market has grown a lot.

These have been the main research projects conducted in this area in India. There is a lack of evaluation and comparison of Online stock trading platforms in India, be it websites or Apps. There is some research regarding some other developing countries like the Philippines, Estonia, etc.

In the Philippines, [8] studied major online stock trading platforms on various criteria such as cost incurred directly or indirectly, reliability and security of the platform, features, ease of use, and perceived customer service. It was found that the traders in the Philippines were highly satisfied with the reduction in cost incurred due to online trading systems. It was also found that they give the most important to the ‘reliability’ of the platform when choosing which platform to use.

The findings from the literature review suggest that there is a gap concerning the investors’ or consumers’ behavior on online stock trading ‘Apps.’ The convenience of mobile applications, along with the rise in popularity of mobiles and the internet, make stock trading platforms readily available to many Indian, especially millennials who have just started working/ have extra income [9]. The study of Indian investors’ behavior and comparison has not been conducted previously [1]. In 2020, there were a lot of stock trading brokers who were available online and through mobile applications. This warrants research in this area. Further, there has been little study of different applications of stock trading (Zerodha, Motilal Oswal Angel brokers, ICICI Securities, etc.) with respect to customer expectations. This area also needs to be looked at.

VI. RESEARCH METHODOLOGY

- Research Type – Descriptive Research
- Sample Size – 40
- Sampling Area - Nagpur
- Sampling Method – Snowball Sampling
- Data Collection Method – Through Questionnaire
- Data Analysis Method – Excel and SPSS

VII. DATA ANALYSIS

A. Descriptive Analysis (Demographic Data)

- Gender Distribution of Respondents

From figure 1, we can infer that 60% of the total respondents were males and the remaining 40% females.

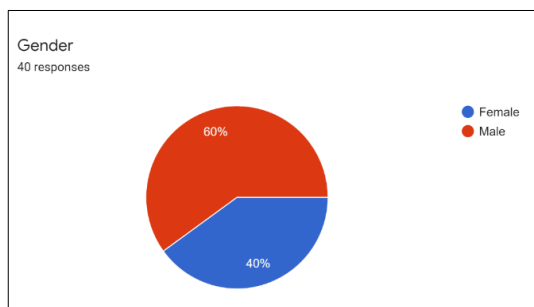


Fig. 1. Gender Distribution of Respondents

• Age Distribution of Respondents

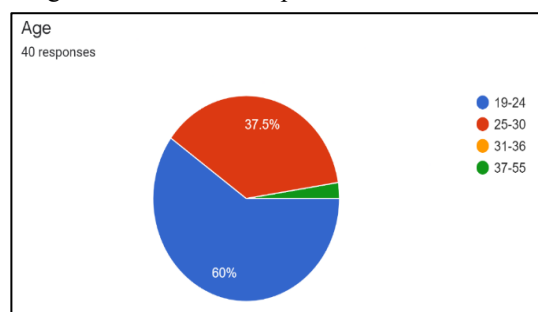


Fig. 2. Age Distribution of Respondents

From figure 2, we can infer that 60% of the total respondents belonged to the 19 to 24 years of age group, 38% of them belonged to 25 to 30, and the remaining belonged to the age group of 37 and above.

• Highest Educational Qualification

From figure 3, we can infer that 60% of the respondents were post-graduates, 37.5% Graduates, and remaining High school diploma holders.

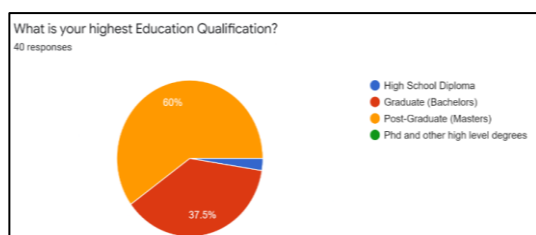


Fig. 3. Highest Education Qualification

• Occupation

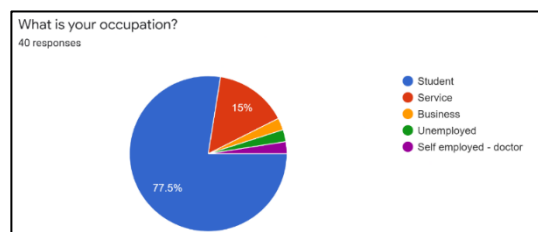


Fig. 4. Occupation.

From figure 4, we can infer that 77.5% of the respondents are students and 15% are service. Rest is insignificant.

• Income

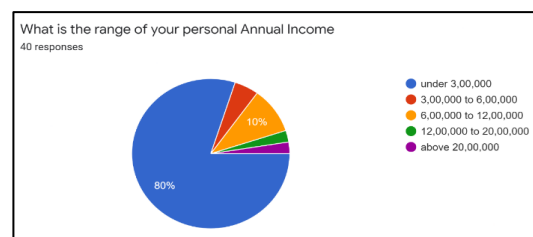


Fig. 5. Income

We can infer from the above figure that 80% of respondents had under three lakhs income. 10% had six lakhs to 12 lakhs income.

B. Reliability Analysis

Cronbach's Alpha = 0.931

The scale used according to Cronbach's Alpha Reliability Statistics is reliable.

C. Hypothesis Testing

- H0: There is no significant difference in the number of Indian Millennials consumers of the stock market using online trading apps, websites, and offline
- H1: There is a significant difference in the number of Indian Millennials consumers of the stock market using online trading apps, websites, and offline.

TABLE I. BROKER DATA

		Broker			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	12	32.4	32.4	32.4
	1	22	59.5	59.5	91.9
	2	3	8.1	8.1	100.0
	Total	37	100.0	100.0	

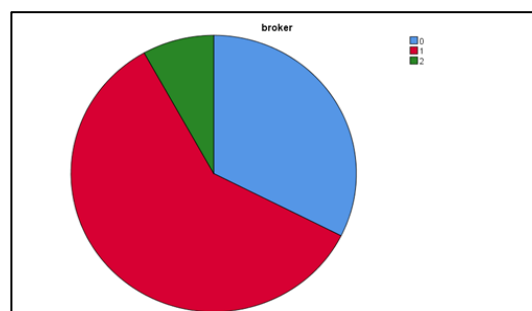


Fig. 6. Broker Data

Here, 0 represents not being aware at all, 1 – aware, 2 – used before, 3 – currently using.

As per the responses, 60% are just aware of the stock trading through brokers. Only three respondents used stock trading through a broker before, and no one is currently using stock trading through a broker.

➤ Website

TABLE II. WEBSITE DATA

Website					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	9	24.3	24.3	24.3
	1	18	48.6	48.6	73
	2	1	2.7	2.7	75.7
	3	9	24.3	24.3	100
	Total	37	100	100	

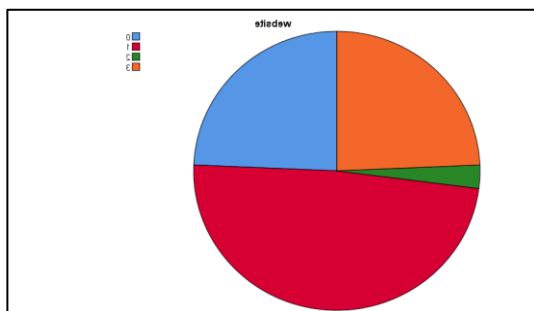


Fig. 7. Website Data

In the case of stock trading through the website, only 24% were unaware of it. 76% of users are aware – of which a significant number approx. 50% are currently using websites for stock trading.

➤ App

TABLE III. APP DATA

		Frequency	Percent	Valid Percent	Cumulative Percent
valid	0	4	10.8%	10.8%	10.8%
	1	14	37.8%	37.8%	48.6%
	2	6	16.2%	16.2%	64.9%
	3	13	35.1%	35.1%	100.0%
	Total	37			

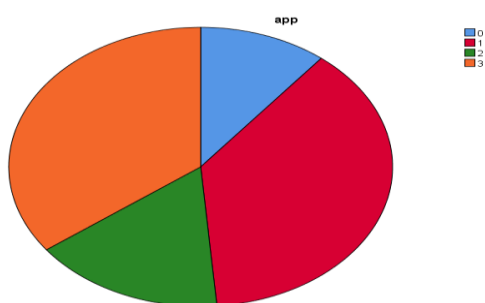


Fig. 8. App Data

Awareness of stock trading apps has reached the max – 90% of users are aware of it. And 35% (max amongst apps, websites, and brokers) are currently using it. Applying correlation between the number of App, website, and offline users, we get mean as 0.273, 0.497, and 0.369, respectively.

Therefore, we can say that there is a significant difference between them.

Therefore, the null hypothesis is rejected.

There is a significant difference in the number of Indian Millennials consumers of the stock market using online trading apps, websites, and offline.

- H0: There is no significant difference in the number of trading actions performed per consumer between trading app consumers, website consumers, and offline consumers

H1: There is a significant difference in the number of trading actions performed per consumer between trading app consumers, website consumers, and offline consumers

Applying correlation between the number of App, website, and offline users, we get mean as 0.239, 0.269, and 0.127, respectively. Therefore, we can say that there is no significant difference between them.

Therefore, the null hypothesis is accepted.

There is no significant difference in the number of trading actions performed per consumer between trading app consumers, website consumers, and offline consumers.

- H0: There is no significant difference in the amount of money invested per consumer between trading app consumers, website consumers, and offline consumers
- H1: There is a significant difference in the amount of money invested per consumer between trading app consumers, website consumers, and offline consumers

Applying correlation between the number of App, website, and offline users, we get mean as 0.162, 0.520, and 0.193, respectively. Therefore, we can say that there is a significant difference between them.

Therefore, the null hypothesis is Rejected.

There is a significant difference in the amount of money invested per consumer between trading apps, websites, and offline consumers.

VIII. FINDINGS

Mostly, people prefer apps over websites and brokers. Amongst the millennials, no one is currently using any brokers (amongst the participants). In general, people recognize many advantages of apps over brokers and websites. Website users are not diminished, though. A significant percentage still use websites, even though they prefer apps.

Many users of Zerodha are present, even though angel broking received the highest NPS and overall rating. There is a significant difference between the number of consumers using the App, websites, and brokers. Most of the millennials in Nagpur prefer App, then website, then broker. The trading actions performed per consumer in apps significantly vary with the trading actions performed per consumer using Brokers (offline). There is a significant difference in the consumer behavior of Nagpur Millennials between broker and app consumers regarding money invested, trading actions performed, and the amount of money invested.

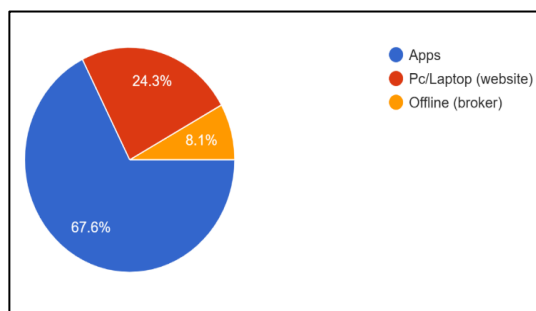


Fig. 9. Preference for Stock Trading format

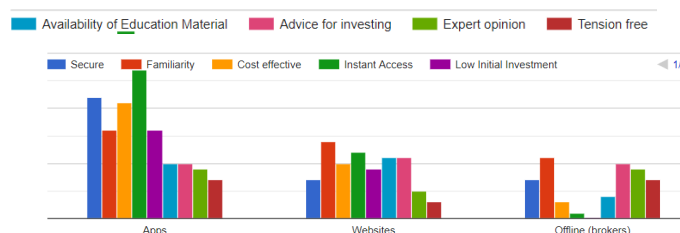


Fig. 10. Advantages of Apps, websites, and brokers

Apps are the most preferred – the main reason for the being Instant Access. Secure and cost-effective transactions are also important. Respondents were asked to rate the importance of Parameters such as Ease of Trading, ease of Understanding Information, the popularity of the App, Ease of Access, High Security, Low brokerage and other charges, AI Interface, Graphics, and UI/UX. 81% thought the ease of trading to be the most important parameter while selecting an app. 72 to 78% of Users thought the ease of understanding information, high security, AI Interface, and UI/UX to be the most critical parameters. Only 50% to 60% of the users considered parameters such as the popularity of the App, low brokerage charges, graphics, and ease of access to be important.

IX. LIMITATIONS & FUTURE SCOPE

The number of respondents surveyed is quite less as the number of people trading and investing in the millennial age group in Nagpur is relatively less. It is also Sensitive Data, so many people do not want to share it. The number of stock trading apps in India is many. Except for Zerodha, the number of users for the rest of the apps is relatively low. Not all the respondents were users of the apps mentioned. Therefore, it was not possible to conclude the competitor analysis.

Further, expanding this research would require combining data from NDSL and CDSL that will give a better picture of consumers' changing preferences and patterns due to the availability of many apps.

X. SUGGESTIONS AND RECOMMENDATIONS

Direct marketing and inorganic advertisement could help UX and UI performers such as Angel Broking and Motilal Oswal. Other apps such as Groww, Upstox, 5paisa, and even Zerodha need to increase the UI and UX performance. The Stockbrokers must note the behavior adopted by Indian/Nagpurian Millennials, as this age group will be quite influenced in the future. Familiarity with the apps will make the websites irrelevant in the future, so Stock traders should focus on improving apps, as currently, the main reason consumers use websites is their familiarity. For Apps, the focus should be given to ease of trading. Followed by the ease of understanding information, high security, AI Interface, and UI/UX.

XI. CONCLUSIONS

Apps are most preferred amongst the millennial traders of Nagpur. Zerodha has captured the maximum market share, while the rest do not have any advantage over the others in terms of market capture. Brand awareness is not lacking amongst the other stock trading Apps, but the conversion/adaptability rate is very low. As the familiarity of the stock trading apps grows, the number of users will shift from websites and brokers.

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